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- [Evaluation](#)
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- [Evaluation Reports](#)
 - [Subject Scores Report](#)
- [Reports](#)
 - [Exam - Question Paper](#)

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*Select Department MSIS

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*Select Topic MSIS - Midterm Examination

Select Difficulty Level --Select--

Select Question Type --Select--

Suggested Mark

Question Text

Search Clear Search

S.No	Syllabus Name	Topic Name	Ques. Type	Diff. Level	Marks	Question	Action
1	BDA 5202	MSIS - Midterm Examination - Mar 2025	Descriptive	Medium	10	A group messaging system allows members of a group to exchange text messages. The system does not	Edit

limit the number of groups that exist in the system.

However, the system is required to enforce the following constraints:

(C1) A group must have a unique name in the entire system.

(C2) A group can have at most 200 members.

(C3) A user can create at most 10 groups.

(C4) A user can be a member of at most 16 groups.

(C5) A group can have one or more admins.

(C6) There is no limit on the number of admins for a group.

(C7) Each text message can be at most 512 bytes in size.

(C8) A response message may refer to a prior message.

(C9) There are two types of messages: Simple and Critical. Create a set of syntactically correct definitions for the necessary collections and documents using MongoDB's schema specification language.

2	BDA 5202	MSIS - Midterm Examination - Mar 2025	Descriptive	Medium	10	Given the group messaging system you described in question (1) above, justify the readConcern and writeConcern you may want to use while performing the following operations: (a) making an existing member the	Edit
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						admin of the group.	
						(b) reading messages from a group.	
						(c) sending a critical message to a group.	
						(d) deleting a message for the entire group.	
						(e) deleting a group for which you are an admin.	
						Differentiate between the following concepts:	
						(a) Replica Set and Sharded Cluster in MongoDB architecture.	Edit
3	BDA 5202	MSIS - Midterm Examination - Mar 2025	Descriptive	Medium	10	(b) Eventual consistency guarantee versus ACID guarantee.	
4	BDA 5202	MSIS - Midterm Examination - Mar 2025	Descriptive	Medium	10	For the problem statement in question (1) above, define SQL tables, indices, and constraints	Edit

						that you may require to support a fully functional messaging system.
						State the importance of the following principles and/or ideas:
5	BDA 5202	MSIS - Midterm Examination - Mar 2025	Descriptive	Medium	10	(a) CAP theorem (b) BASE transaction model (as against ACID model)
						Edit